DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2021/2022

THIRD YEAR **SEMESTER II** EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CIT 2207: Mobile Computing

DATE: **TIME:**

Instructions: Answer Question 1 and Any Other Two

Question 1: (30 Marks)

a) Define the following terms (3 marks)
   i. Session Mobility
   ii. Service Mobility
   iii. Polling

b) Differentiate between *mobility* and *wireless* terminologies as used in mobile computing. (2 marks)

c) Explain the role of Wireless Markup Language (WML) in wireless applications. (2 marks)

d) Assume that your friend recently travelled to West Africa and you just made a phone call to her. She is surprised and is asking how her Safaricom Kenya line worked in Ghana. Discuss mobile IP topology and explain how it facilitates such kind mobility management beyond a particular location area (4 marks)

e) Briefly discuss three strategies used for handoff detection (3 mark)

f) Explain three reasons for a handoff to be conducted in mobility management (3 marks)

g) Explain any TWO problems associated with Mobile IP (2 marks)

h) Describe the components of the wireless application protocol (WAP) architecture (5 marks)

i) Many a times we visit overcrowded places such as graduations ceremonies for university students or even when police officers are graduating we have had cases where by family, relative and friends of individual graduating fail to meet the graduand or even each other because during that time calls fails to go through, sometime calls terminates and fail to connect again. To reduce force termination and promote call completion channel assignment scheme have been proposed. Discuss four channel assignment schemes that have been proposed (4 marks)

j) In mobile computing, future developments will definitely provide new technologies and better computing concepts. Explain any TWO of issues being addressed by these (2 marks)

k)

Question 2: (20 Marks)

a) Discuss four types of handover (handoff) managements used in mobile cellular technologies (8 marks)
b) Explain the steps involved in the call delivery procedure in GSM network in the following cases: (6 marks)
   i) GSM mobile terminated call
   ii) GSM mobile originated call

c) John Travelled to France for a holiday. While in France John was able to check balance on his mobile phone, he could also receive calls and short messages as well as send and call his family back home in Kenya using his Safaricom Line. This can be explained using roaming management in mobility; thus, one is able to make calls or even receive messages while outside his or her geographical region. Using a diagram discuss the three key components of signaling system 7 (SS7) network that necessitated John to make and receive calls and messages while in France. (6 marks)

Question 3: (20 Marks)

a) Mobile communication is resource constrained. Therefore, it requires special operating environment which translates TCP/IP protocols into an equivalent of wireless applications environment. Using a model diagram, discuss how the WAP architecture seamlessly integrates www internet applications (6 marks)
b) Using a diagram, describe the IP mobile registration process indicating clearly all entities involved in registration process. (4 marks)

c) Briefly explains the improvements of mobile IPV6 that has solved the challenges raised in Mobile IPV4. (2 marks)

d) Describe in detail the following occurrence of media access in mobile computing. Use diagram where necessary.
   i. Exposed Terminals (2 marks)
   ii. Hidden Terminal (2 marks)
   iii. Near and Far Terminal (2 marks)

Question 4: (20 Marks)

a) Explain roaming management and its two basic operations as used in mobility management (3 marks)
b) Explain how Frequency reuse help channel allocation of cellular systems (3 marks)
c) Explain FOUR challenges of mobile computing (4 marks)
d) Explain four dimensions in which multiplexing can be carried out in wireless communication. (8 mark)
e) Explain mobility binding in mobile ip process (2 marks)

END.